II TEST - 2026-27

Blueprint

Chapter	Teaching Hours	Marks Weightage	Part A 1mark 10 Qs (10 Marks)	Part B 2mark 5 Qs (10 Marks)	Part C 3mark 5 Qs (15 Marks)	Part D 5mark 4 Qs (20 Marks)	Total Questions
11. Photosynthesis in Higher Plants	7	8	3	1	1	*	5
12. Respiration in Plants	7	8	1	1	*	1	3
13. Plant Growth and Development	8	9	1	*	1	1	3
14. Breathing and Exchange of Gases	6	7	2	1	1	*	4
15. Body Fluids and Circulation	7	8	*	*	1	1	2
16. Excretory Products and their Elimination	6	7	2	1	1	*	4
17. Locomotion and Movement	7	8	1	1	*	1	3
Total	**48**	**55**	**10**	**3/5**	**3/5**	**2/4**	**18/24**

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MODEL PAPER

General Instructions:

1. The question paper consists of Parts, A, B, C and D
2. Part-A consists of I and II
3. All the parts are compulsory.
4. For Part-A questions, only the answers written first will be considered for evaluation.
5. Draw diagrams wherever necessary. Unlabelled diagrams do not attract any marks.

PART – A

I. Select the correct alternative from the choices given below: $7 \times 1 = 7$

1. Read the following four statements, (i), (ii), (iii) and (iv) and select the right option having both correct statements.

   (i) Z scheme of light reaction takes place in presence of PSI only.

   (ii) Only PSI is functional in cyclic photophosphorylation.

   (iii) Cyclic photophosphorylation results into synthesis of ATP and NADPH2.

   (iv) Stroma lamellae lack PSII as well as NADP.

   a) (ii) and (iv) b) (i) and (ii) c) (ii) and (iii) d) (iii) and (iv)

2. With reference to factors affecting the rate of photosynthesis, which of the following statements is not correct?

   a) Increasing atmospheric CO_2 concentration up to 0.05% can enhance CO_2 fixation rate.

   b) C_3 plants respond to higher temperature with enhanced photosynthesis while C_4 plants have much lower temperature optimum.

   c) Tomato is a greenhouse crop which can be grown in CO_2-enriched atmosphere for higher yield.

   d) Light saturation for CO_2 fixation occurs at 10% of full sunlight.

3. Match the list-I with list-II.

List- I	List-II
A. Molecular oxygen	i. α-Ketoglutaric acid
B. Electron acceptor	ii. Hydrogen acceptor
C. Pyruvate dehydrogenase	iii. Cytochrome C
D. Decarboxylation	iv. Acetyl Co A

Choose the correct answer from the options given below:

a) A- ii B-iv C-iii D-I b) A- iii B-i C-ii D-iv

c) A- iiB-iv C-i D-iii d) A- iii B-iv C-ii D-I

4. Spraying sugarcane crop with a plant hormone increases the length of the stem, thus increasing the yield by as much 20 tonnes per acre. The plant hormone is

   a) Gibberellin b) Auxin c) Cytokinin d) Abscisic acid

5. Emphysema is a chronic respiratory disorder, in which the organ affected is

   a) Alveolar walls b) Bronchi c) Bronchioles d) Pleural membrane

6. Statement 1: Glucose & amino acids are re-absorbed by active process.

   Statement 2: Re-absorption of glucose and ammonia does not require energy.

   a) Statement I & statement II are true b) Statement I is true & statement II is false

   c) Statement I & statement II are false d) Statement I is false & statement II is true

7. Choose the wrongly matched pair.

   a) Isotropic band - Actin

   b) Anisotropic band - Myosin

   c) Central part of I- band - M-line

   d) Central part of A-band - H-zone

II. Fill in the blanks by choosing the appropriate word from those given in the bracket: **$3 \times 1 = 3$**

(Residual, 700nm, renal calculi, 680nm)

8. The reaction centre in PSII has an absorption maximum at _____.
9. The volume of air remaining in the lungs even after a forcible expiration_____volume
10. Insoluble mass of crystalized salts formed within the kidney is _____.

PART – B

III. Answer any THREE of the following questions in 2 to 3 sentences each, wherever applicable: **$3 \times 2 = 6$**

11. "Photorespiration is a wasteful process." Justify.
12. Differentiate alcoholic fermentation from lactic acid fermentation.
13. Give the graphical representation of oxygen dissociation curve.
14. Mention the two effects of aldosterone on nephron.
15. Draw a neat labelled diagram of myosin monomer.

PART – C

IV. Answer any THREE of the following questions in about 40-80 words each, wherever applicable **$3 \times 3 = 9$**

16. Give the schematic representation of the cyclic photophosphorylation.
17. Graphically represent the sequence of the developmental process in a plant cell.
18. Explain the role of diaphragm and rib cage in the breathing process.
19. Study the blood report of subject A, B & C and draw conclusions.

    A) Had high Eosinophil count. B) 3 million RBC/ mm^3 C) 80,000 platelets/ m
20. In regulatory role in kidney function where fall in GFR activates J G cells release A, which converts inactive B in blood to active B and further to C. Identify



PART – D

V. Answer any TWO of the following questions in 200–250 words each, wherever applicable.

$2 \times 5 = 10$

21. Give the schematic representation of glycolysis.
22. Growth hormones exhibit various physiological effects on different plants. In this view, a table is given below, comprising growth hormones, their physiological effects and crop plant on which they show that effect. Complete the table in this regard.

Auxins	A	Tea plantation
Ethylene	Synchronizing fruit set	B
C	Increase in stem length	Sugarcane
Ethephon	D	Cucumber
E	Flowering	Pineapple

23. Draw a neat labelled diagram of section of human heart.
24. Name the joints in / between the following:

    a) Between carpals

    b) Atlas and Axis

    c) Adjacent vertebrae

    d) Humerus & pectoral girdle

    e) Skull bones

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